

# AYAAN SANADI

Innovative Computer Vision and Machine Learning Engineer specializing in Edge AI, TinyML, and real-time deep learning pipelines. Proven track record of architecting high-performance, low-latency models and full-stack AI systems deployed across resource-constrained hardware and cloud environments.

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## CORE TECHNICAL SKILLS

Python, JavaScript, C, C++, YOLO, MediaPipe, TensorFlow Lite (Micro), PyTorch, FastAPI, OpenCV, Next.js, Git, PostgreSQL, MongoDB

## PROFESSIONAL EXPERIENCE

### Computer Vision Intern

Jan 2026 – Apr 2026

*Neuronetic Vision*

- Engineered end-to-end object detection pipelines using Git for version control, scraping and annotating complex datasets to train YOLO models, boosting data quality by **[Insert % Data Quality Improvement]**.
- Fine-tuned and deployed PyTorch-based YOLO models for real-time CCTV security, achieving **[Insert % Accuracy Improvement]** accuracy in detecting weapons and public violations.
- Architected automated warehouse vision solutions for pest and license plate detection, reducing manual monitoring latency by **[Insert Latency Reduction %]** under varied lighting.

## KEY ENGINEERING PROJECTS

### FormCheck AI: Real-Time Biomechanics Analysis

- Architected an AI gym safety platform using MediaPipe and OpenCV for real-time pose estimation, achieving **[Insert Frame Rate in FPS]** FPS processing speeds on standard hardware.
- Engineered a custom validation pipeline to deliver instant corrective audio cues, reducing form-tracking latency by **[Insert Latency Reduction in ms]** ms and storing telemetry in MongoDB.

### BirdSpec: End-to-End Species Detection Pipeline

- Trained a lightweight EfficientNet-V2 CNN in PyTorch on 60,000+ images to classify 483 bird species, achieving **[Insert % Accuracy]** accuracy despite high inter-class similarity.
- Deployed the inference engine via FastAPI, integrating a Next.js frontend and MongoDB backend to reduce history-tracking query latency by **[Insert MongoDB Query Speedup %]**.

### Real-Time AI Smart Attendance & Analytics Pipeline

- Automated attendance for 70+ students by engineering a YOLOv8-face and InceptionResnetV1 pipeline, achieving **[Insert % Accuracy Improvement]** accuracy in facial feature extraction.
- Architected a 'Two-Brain' tracking system pairing DeepOCSORT with long-term logical memory, maintaining **[Insert Target FPS]** FPS and eliminating ID-switching errors.
- Accelerated processing by offloading tensor operations to Apple Silicon (MPS), reducing frame matching latency by **[Insert Latency Reduction %]** via vectorized cosine similarity.
- Integrated a PostgreSQL backend to translate AI outputs into official registry data, reducing administrative reporting overhead by **[Insert % Time Saved]** through automated CSVs.

### AI-Powered Acoustic Pest Detection System (Edge AI / TinyML)

- Deployed an INT8 post-training quantized CNN with a sub-65KB footprint onto an ESP32 using TFLite Micro, achieving **[Insert % Accuracy]** accuracy in real-time pest detection.
- Built a custom C/C++ DSP pipeline to replicate Python's Librosa at the edge, converting raw 16kHz audio to Mel-Spectrograms and reducing processing latency by **[Insert Latency Reduction %]**.
- Prevented ESP32 RAM overflow and tensor corruption by forcing 16-byte memory alignment in C and shifting audio arrays to the Heap, ensuring **[Insert % Uptime]** system stability.
- Designed a 5-step rolling buffer with '3-out-of-5' voting logic, reducing false-positive alerts by **[Insert % Reduction]** and pushing instant HTTP POST alerts to Next.js.

## EDUCATION

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**BMS Institute of Technology and Management**

2024 – 2028

*B.E. in Artificial Intelligence & Machine Learning | CGPA: 8.6*